

# Missing Data

Every data set in this book so far has arrived complete. Real ones do not. A respondent skips the question about income, a patient stops coming to the clinic, a sensor fails for three weeks, a laboratory loses a sample. The holes are rarely scattered at random, and the software's silent default — drop every record with a hole in it — is a decision, not a neutral act.

This chapter treats the holes as data. The pattern of what is missing has a distribution of its own, and the subject turns on what that distribution may depend on. If it depends on nothing, almost any analysis survives. If it depends only on what was recorded, likelihood and Bayesian inference survive and most shortcuts do not. If it depends on what was *not* recorded, nothing in the observed data can tell you so. Those three cases organize what follows: when dropping incomplete records is safe, why single imputation lies about precision, what EM is and what it does not give you, how multiple imputation restores the uncertainty imputation destroys, and how to report a conclusion resting on an untestable assumption.

**What you need.** [Chapter 5](#) and [Chapter 6](#) (the linear model and least squares), [Section 7.3](#) (maximum likelihood Theorem 7.3.1, the score and information Proposition 7.4.7), [Section 7.6](#) and [Section 12.6](#) (the conjugate Bayesian linear model Theorem 7.6.2 and credible sets Definition 12.6.1), [Chapter 11](#) (the likelihood ratio test Theorem 11.4.1). [Section 18.6](#) (Theorem 18.6.1, filling in a lost plot of a designed experiment) is the special case this chapter generalizes. [Chapter 23](#) supplies the bootstrap (Definition 23.1.1), which is one of the two honest routes to a standard error after EM; [Chapter 25](#) the habit of asking what a coefficient is meant to answer; and [Chapter 34](#) the generalized linear model (Definition 34.2.1, Theorem 34.3.1), since nothing here is special to a normal response. [Section 41.4](#) works out the EM algorithm for a normal response, and for that it needs the conditional distribution of one normal block given another (Theorem 3.5.2, [Chapter 3](#)). Only three earlier results are cited inside a proof here: that one, Theorem 7.3.1 in [Section 41.3](#), and Theorem 18.6.1 in [Section 41.4](#). The rest of the list is a place to go back to — the linear model and its likelihood, the tests and credible sets a completed data set would be analysed with, and the bootstrap and the causal question of Chapters 23 and 25.

## 41.-1.1 Roadmap

- [Missingness mechanisms](#)
- [Complete cases and quick repairs](#)
- [Likelihood under missingness at random](#)
- [The EM algorithm](#)
- [Multiple imputation](#)
- [Sensitivity analysis](#)
- [Summary and notes](#)